Exterior Robin Certificates for Electromagnetic Tails

A source-specific Ram Calculus audit of manuscript M514

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Abstract

We audit the pinned manuscript *Electromagnetic tails and the Kerr–Newman Penrose inequality*, retaining its exterior domain, Newton normalization, minimal-sphere Robin condition, and bare gravitational angular momentum. We prove explicit weighted bounds for its positive Kelvin-reflected inverse; specialize nonlinear residual, perturbation, and truncation certificates to its two negative-power sources; and propagate certified state errors to mass, boundary area, and the inequality defect. Compactness and an elementary energy argument extend scalar existence and uniqueness to every finite amplitude without a contraction hypothesis. Explicit derivative estimates make the radial geometric barrier quantitative, and an exact-rational finite-parameter test gives a conservative negative-defect certificate, conditional on the audited seed-to-geometry identities. A separate tail analysis explains the possible logarithmic dipole remainder and proves a nonzero subleading logarithmic quadrupole in the actual source family; the exact mass-flux argument avoids any finite-first-moment assumption. Classical potential theory and contraction arguments are distinguished from their private Ram representation and validation contracts. The source theorem as a whole is not claimed independently verified, and neither a new total-angular-momentum counterexample nor a measured computational speedup is asserted. Contribution assessment: C4/5, removal of scalar smallness restrictions together with a proved resonant tail and quantitative geometry-to-defect certificates gives a substantial local extension while arbitrary-amplitude horizon geometry and independent numerical PDE validation remain unproved.

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1 Pinned Problem and Exact Conventions

1.1 Identity, scope, and provenance

The target is catalogue M514, family 260, applicability L5, at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Its source date is October 5, 2026; this report and public-literature check are dated October 7, 2026. L5 denotes relevance, not contribution or correctness. The source root relative to the study root is

```
_sources/openai-math/preprints/Electromagnetic-tails-and-the-Kerr-Newman-Penrose-inequality-October-5-2026/
```

Every substantive source section was read, including the full geometric construction, all elliptic proofs, all flux calculations, and the equality argument. The master file includes precisely Sections 00–04 and a bibliography; it has no mathematical appendix. Read scope and SHA-256 input hashes are recorded in the author audit. Local confidential corpus text was not used in public searches or uploaded.

Here "proved locally" means a proof supplied in this report under its displayed hypotheses. "Source dependency" means a step read and audited but not promoted to an independently verified global theorem. Symbolic and exact-rational checks test identities and inequalities, not every analytic or geometric hypothesis.

1.2 The geometric assertion actually under consideration

Use $G = c = 1$, zero cosmological constant, and the oriented exterior

$$\Omega = \{x \in \mathbb{R}^3 : r = |x| \geq 1\}, \qquad S = \{r = 1\}.$$

For a positive conformal factor u, covariant symmetric tensor σ, and Euclidean vector fields $\widetilde{E}, \widetilde{B}$, set

$$g = u^4\delta, \quad K = u^{-2}\sigma, \quad \mathcal{E} = u^{-6}\widetilde{E}, \quad \mathcal{B} = u^{-6}\widetilde{B}, \quad \tau = \operatorname{tr}_g K = 0. \tag{1.1}$$

The cross-product sign is $(X \times_g Y)^\flat(Z) = \mathrm{d}V_g(X, Y, Z)$. The equations are

$$\begin{aligned} R_g - |K|_g^2 &= 2(|\mathcal{E}|_g^2 + |\mathcal{B}|_g^2), & \operatorname{div}_g K &= 2(\mathcal{E} \times_g \mathcal{B})^\flat, \\ \operatorname{div}_g \mathcal{E} &= 0, & \operatorname{div}_g \mathcal{B} &= 0. \end{aligned} \tag{1.2}$$

For the normal toward the asymptotic end, $H = \operatorname{div}_\Sigma \nu$ and $\theta_+ = H + \operatorname{tr}_\Sigma K$. The boundary is required to have $\theta_+ = 0$. Outermostness excludes every compact smooth embedded enclosing surface in the interior, possibly disconnected, with $\theta_+ \le 0$ everywhere. An enclosing cut is the full compact smooth intrinsic boundary $\Gamma = \partial D$ of a connected smooth codimension-zero $D \subset \Omega$, closed in Ω, containing the whole distant end, with interior in $\operatorname{int}\Omega$. All components and all portions coinciding with S count in its area. The assertion is $\operatorname{Area}_g(\Gamma) \ge A = \operatorname{Area}_g(S)$ for *every* such cut, not only axisymmetric cuts or radial graphs.

The end has Cartesian falloff $g - \delta = O_4(r^{-1})$, $K = O_3(r^{-3})$, and $\mathcal{E}, \mathcal{B} = O_3(r^{-2})$, where each derivative lowers the power by its order. The total constraint energy and momentum densities are integrable. The effective $U(1)$ generator $\eta = \partial_\phi$ has period 2π. In this family $P_{\mathrm{ADM}} = 0$, and

$$\begin{aligned} m &= -\frac{1}{2\pi} \lim_{R\to\infty} \int_{S_R} u^3 u_r \, \mathrm{d}A_\delta, \\ J &= \frac{1}{8\pi} \lim_{R\to\infty} \int_{S_R} K(\nu_R, \eta) \, \mathrm{d}A_g, \qquad Q_{e,b} = \frac{1}{4\pi} \lim_{R\to\infty} \int_{S_R} \iota_{\mathcal{E},\mathcal{B}} \, \mathrm{d}V_g. \end{aligned} \tag{1.3}$$

The source concerns

$$m^2 \ge \frac{A}{16\pi} + \frac{Q^2}{2} + \frac{\pi(Q^4 + 4J^2)}{A}, \qquad A \ge 4\pi\sqrt{Q^4 + 4J^2}, \quad Q^2 = Q_e^2 + Q_b^2, \tag{1.4}$$

with *bare gravitational* J as in (1.3). It does not refute the assertion using conserved total angular momentum.

1.3 The exact scalar problem

Fix a smooth $\chi : \mathbb{R} \to [0,1]$, zero on $(-\infty, 1]$ and one on $[2, \infty)$. Let $L > 4$, $s \in [0,1]$, $c(r) = \chi(r/L)$, and $t = \cos\theta$. The coefficients are

$$\begin{aligned} p &= \frac{s^4 c^4 (1-t^2)^3}{4r^6}, \\ \ell &= \frac{s^2}{4r^4}\{c^2(9t^4 - 2t^2 + 1) + (rc')^2(1 - t^4)\}. \end{aligned} \tag{1.5}$$

They vanish on $r \le L$, are nonnegative and smooth, and obey $D^j p = O(s^4 r^{-6-j})$, $D^j \ell = O(s^2 r^{-4-j})$. We solve, pointwise after bootstrap,

$$-\Delta u = p u^{-7} + \ell u^{-3}, \qquad u_r + \tfrac{1}{2} u = 0 \ (r = 1), \qquad u \to 1 \ (r \to \infty). \tag{1.6}$$

The derivative ∂_r points *into* Ω at its inner boundary; the outward domain normal is $-\partial_r$. In outward-normal notation the homogeneous Robin coefficient is therefore $-1/2$. The lift $u_0 = 1 + r^{-1}$ satisfies both homogeneous Laplace and Robin conditions and has the required limit. Our Banach space is the real $X = C_b(\Omega)$ with sup norm; its complete nonnegative cone is X_+. For source estimates use

$$Y_{k,L} = \{H \in C(\mathbb{R}^3) : H = 0 \text{ on } r < L, \ \|H\|_{k,L} := \sup_{r \ge L} r^k |H| < \infty\}.$$

The actual sources vanish smoothly near $r = L$ as required by the cutoff. Measurable radial envelopes used below need not themselves be continuous.

2 Stepwise Source Dependency Audit

2.1 Seeds and conformal reduction: source Section 01

The source defines $f = sc\sin^2\theta$, $h = sc\sin^2\theta\cos\theta$, $\iota_{\widetilde{E}}\,\mathrm{d}V_\delta = \mathrm{d}(f\,\mathrm{d}\phi)$, $\iota_{\widetilde{B}}\,\mathrm{d}V_\delta = \mathrm{d}(h\,\mathrm{d}\phi)$, and

$$\sigma = -s^2c^2r^{-2}\sin^4\theta(\,\mathrm{d}r\otimes\,\mathrm{d}\phi + \,\mathrm{d}\phi\otimes\,\mathrm{d}r).$$

In `lem:seeds`, the Cartesian primitives $scr^{-2}(x^1\,\mathrm{d}x^2 - x^2\,\mathrm{d}x^1)$ and $scx^3r^{-3}(x^1\,\mathrm{d}x^2 - x^2\,\mathrm{d}x^1)$ remove apparent axis singularities. Exactness gives both divergence constraints and zero charges. The coordinate components are

$$(\widetilde{E}^r, \widetilde{E}^\theta) = (2sct/r^2, -sc'\sin\theta/r^2), \quad (\widetilde{B}^r, \widetilde{B}^\theta) = (sc(3t^2-1)/r^2, -sc'\sin\theta t/r^2).$$

The polar components are coordinate components: multiply by r to obtain orthonormal components. The source checks all tensor-divergence components, with the only nonzero one $\partial_r\sigma_{r\phi} + 2\sigma_{r\phi}/r = -2s^2cc'\sin^4\theta/r^2$. It equals twice the electromagnetic cross-product covector component and is supported on $L \le r \le 2L$.

In `lem:conformal`, $\mathrm{d}V_g = u^6\,\mathrm{d}x$ preserves Maxwell flux forms, the tracefree covariant tensor divergence scales by u^{-6}, and $R_g = -8u^{-5}\Delta u$. Squared norms scale by u^{-12} for K and u^{-8} for the vector fields. These identities yield exactly $p = |\sigma|^2/8$ and $\ell = (|\widetilde{E}|^2 + |\widetilde{B}|^2)/4$; neither an extra 4π nor a sign change is permitted. This reduction, including smoothness on the axis and conformal tensor types, remains a geometric input to our local scalar results.

2.2 Elliptic chain: source Section 02

`lem:newton` proves existence and C^1 regularity of the Newton potential, source-separated higher derivatives, Kelvin reflection, its Robin trace, positivity, and uniqueness of the decaying homogeneous problem. `lem:regularity` transfers all but one derivative to a localized source, retaining only a locally integrable first derivative of the Newton kernel. It does not differentiate the singular kernel arbitrarily often under an unlocalized integral. This is crucial.

`prop:conformal-factor` contracts the whole nonnegative C_b cone with an unspecified CL^{-2} constant, obtains $u_0 \le u \le 3$, then bootstraps the fixed point to all Cartesian derivative orders. The C^1 start, support separation near S, smooth coefficient cutoff, and lower bound $u \ge 1$ are distinct dependencies. Uniqueness there is stated for $v = u - u_0 \ge 0$, $v = O(r^{-1})$. Parameter continuity uses a uniform contraction, then weighted dominated convergence for the source integrals. Sections 3–6 make constants and error propagation explicit and extend the scalar uniqueness class.

2.3 Geometric chain: source Section 03

Uniform comparison $\delta \le g \le M^4\delta$ proves metric completeness with the boundary included. Derivative decay proves the stated end regularity. The energy density is $O(r^{-4})$; the momentum density is compactly supported because the two tail fields are parallel and radial. `lem:positive-spheres` computes

$$W = u + 2ru_r, \qquad \theta_+(S_r) = \frac{2W}{ru^3}. \tag{2.1}$$

Tangential restriction of K to every coordinate sphere vanishes. The Robin condition makes $W(1) = 0$; bounds for v, v_r and v_{rr} yield strict positivity for all $r > 1$, including arbitrarily close to S. At a largest-radius tangency point of any enclosing closed component, $H_\Sigma \ge H_{S_r}$ and $\mathrm{tr}_\Sigma K = 0$. This is the outermostness argument, not a small-norm claim alone. The source also discusses edged components by detouring paths around their edges; the cut definition itself uses the full smooth intrinsic boundary.

`lem:area-minimum` uses the closed form

$$\beta = \pi^*\{u(1,\omega)^4\,\mathrm{d}A_{\mathbb{S}^2}\}, \quad \|\beta\|_{*,g} = \frac{u(1,\omega)^4}{r^2 u(r,\omega)^4} \le 1, \quad \pi(r,\omega) = \omega.$$

The last inequality follows from $\partial_r(ru^2) = uW > 0$. Stokes' theorem on $D \cap \{r \le R\}$ includes *all* components of Γ, including contact with S, and gives $|\int_\Gamma \beta| = A$. This proves the required all-cuts comparison. The topology, orientation, tangency, and Stokes steps have been read; we do not replace them by an elliptic residual certificate.

2.4 Fluxes and final inference: source Section 04

`prop:charges` uses $K = O(r^{-3})$ to give $P_{\mathrm{ADM}} = 0$, exact two-form fluxes to give $Q_e = Q_b = 0$, and Newton divergence flux to give the exact mass identity

$$m = 2 + \frac{1}{2\pi}\int_\Omega (1 + r^{-1})H\,\mathrm{d}x, \qquad H = pu^{-7} + \ell u^{-3}. \tag{2.2}$$

This is $2 + 2\mathcal{N}H(0) + (2\pi)^{-1}\int H$. It does not require $\int |x||H|\,\mathrm{d}x < \infty$. Cancelling the conformal weights in the angular flux gives $J_{\mathrm{grav}}(r) = -4s^2c(r)^2/15$. `prop:corrected-flux` proves

$$\mathrm{div}_g\{K(\cdot,\eta)^{\sharp_g} + 2h\mathcal{E}\} = 0, \qquad J_{\mathrm{EM}}(r) = \frac{1}{4\pi}\int_{S_r} h\iota_{\mathcal{E}}\,\mathrm{d}V_g = \frac{4s^2c(r)^2}{15}. \tag{2.3}$$

Indeed Cartan's formula gives $\mathcal{E}(h) = -\,\mathrm{d}V_g(\mathcal{E},\mathcal{B},\eta)$; symmetry of K and the Killing equation supply the opposite divergence. Thus $\mathcal{J} = J_{\mathrm{grav}} + J_{\mathrm{EM}} = 0$ at every radius. The potential one-form is globally smooth; this is not a singular-gauge artifact. Its norm decays as r^{-1} while $h = A_B(\eta)$ can be order one, since $|\eta| = O(r)$.

With $J = -4s^2/15$, $m = 2 + O(s^2/L)$, $A = 64\pi + O(s^2/L^2)$, the bare defect $D = m^2 - A/(16\pi) - 4\pi J^2/A$ tends to $-1/225$ at $s = 1$. The source proves $m - 2 \ge s^2/(3M^3L)$ by integrating the nonzero electric tail. This makes D positive at small nonzero s for fixed large L, and continuity supplies an equality root. The strict branch $A \ge 64\pi > 8\pi|J|$ is uniform. At equality, $m^2 - |J| = (A - 8\pi|J|)^2/(16\pi A) > 0$; a zero-charge Kerr–Newman spacetime has zero Maxwell tensor on every spacelike slice, contrary to the constructed nonzero fields. This last step is an obstruction to an induced-data realization, not a proof of a general rigidity theorem.

3 The Exterior Inverse With Explicit Norms

3.1 Kernel, boundary, and uniqueness

For $H \in Y_{4,L}$ extend by zero as already specified and define

$$\mathcal{N}H(x) = \frac{1}{4\pi}\int_{\mathbb{R}^3} \frac{H(y)}{|x-y|}\,\mathrm{d}y, \quad I(x) = x/r^2, \quad \mathcal{R}H(x) = \mathcal{N}H(x) + r^{-1}\mathcal{N}H(I(x)). \tag{3.1}$$

Equivalently the integration kernel on Ω is

$$G_R(x,y) = \frac{1}{4\pi}\left\{|x-y|^{-1} + (r|I(x)-y|)^{-1}\right\}.$$

It is positive, has the Euclidean volume measure, and is not a probability-normalized averaging kernel.

Lemma 3.1 (Exterior inverse contract)**.** *For $H \in Y_{4,L}$, $\mathcal{R}H \in C^1(\Omega)$, is smooth near S, obeys $-\Delta\mathcal{R}H = H$ distributionally, decays as $O(r^{-1})$, and satisfies $(\partial_r + 1/2)\mathcal{R}H = 0$ on S. It preserves positivity. It is the unique solution with those equation, trace, and decay properties in the class $C^1(\Omega)$ with a distributional Laplacian.*

Proof. The kernels $|z|^{-1}$ and its gradient are locally integrable in three dimensions. Truncating a small singular ball and then sending its radius to zero proves continuity and one derivative, uniformly on compact sets. Integrability of H and source separation provide all far-field and inner-ball differentiations. The fundamental solution identity $-\Delta(4\pi|x|)^{-1} = \delta_0$ gives the equation. Kelvin transformation gives $\Delta[r^{-1}\mathcal{N}H(I(x))] = r^{-5}(\Delta\mathcal{N}H)(I(x)) = 0$ on Ω, because $|I(x)| \le 1 < L$. If $n(t) = \mathcal{N}H(t\omega)$, the reflected expression on a ray is $n(r) + r^{-1}n(1/r)$; its value and radial derivative at 1 are $2n(1)$ and $-n(1)$. This proves the trace with the correct factor $1/2$.

For uniqueness subtract two such solutions to obtain harmonic w of order r^{-1}. Extend to $0 < r < 1$ by $r^{-1}w(I(x))$. The values, tangential derivatives, and radial derivatives match at S, since the interior derivative is $-w - w_r = w_r$. Thus the extension is distributionally harmonic away from zero, bounded at zero, and decays at infinity. The removable singularity theorem and Liouville's theorem, or the annular maximum-principle barrier $C\varepsilon/r + C/R$, show that it vanishes. The decay of the particular solution follows also from the radial envelope below. □

Theorem 3.2 (Exact radial majorants)**.** *Let $k > 3$ and $|H(x)| \le ar^{-k}\mathbf{1}_{r\ge L}$. Then*

$$|\mathcal{R}H(x)| \le a\{n_{k,L}(r) + n_{k,L}(0)/r\}, \tag{3.2}$$

where

$$n_{k,L}(r) = \begin{cases} L^{2-k}/(k-2), & 0 \le r \le L, \\ L^{3-k}/((k-3)r) - r^{2-k}/((k-3)(k-2)), & r \ge L. \end{cases} \tag{3.3}$$

In particular,

$$\|\mathcal{R}\|_{Y_{k,L}\to X} = \frac{2L^{2-k}}{k-2}, \qquad \|\mathcal{R}H\|_\infty \le \frac{a}{L^2} \quad (k=4). \tag{3.4}$$

Proof. Angular integration of the Newton kernel gives $(4\pi)^{-1}\int_{\mathbb{S}^2} |r\omega - \rho\xi|^{-1}\,\mathrm{d}A_\xi = 1/\max(r,\rho)$. Integrating ρ^{2-k} over $[L,\infty)$ gives (3.3). The reflected argument always has radius at most 1, so its envelope is $n_{k,L}(0)/r$. The radial Newton potential is nonincreasing, as follows either by differentiating or by $n' = -r^{-2}\int_L^r \rho^{2-k}\,\mathrm{d}\rho$ for $r > L$. Both summands in (3.2) are nonincreasing for $r \ge 1$; their sum attains its maximum $2n_{k,L}(0)$ at $r = 1$. This proves the upper norm bound. The discontinuous radial envelope attains it in the bounded-measurable source class. Continuous envelopes which turn on between L and $L + \varepsilon$, bounded by r^{-k}, approach it by dominated convergence. Thus the operator norm on the stated continuous class equals the same number, whether or not attained. □

The distinction between equality of the norm and attainment is relevant: the hard cutoff is a comparison function, not the manuscript's smooth physical field. In particular, the constants $1/L^2$ for an r^{-4} source and $1/(2L^4)$ for an r^{-6} source are exact weighted operator norms, not fitted numerical values.

4 Quantitative Nonlinear and Ram Certificates

4.1 Explicit cone contraction

Define

$$d_\chi = \sup_{1\le z\le 2} |z\chi'(z)|, \quad P = s^4/4, \quad E = s^2(8 + d_\chi^2)/4, \tag{4.1}$$

so that $p \le Pr^{-6}\mathbf{1}_{r\ge L}$ and $\ell \le Er^{-4}\mathbf{1}_{r\ge L}$. The polynomial $9t^4 - 2t^2 + 1$ lies in $[0,8]$ for $|t| \le 1$ and $1 - t^4 \le 1$. Put

$$b = \frac{P}{2L^4} + \frac{E}{L^2}, \quad q = \frac{7P}{2L^4} + \frac{3E}{L^2}, \quad T(v) = \mathcal{R}\{p(u_0+v)^{-7} + \ell(u_0+v)^{-3}\}. \tag{4.2}$$

Theorem 4.1 (An explicit continuum certificate)**.** *If $q < 1$, T has exactly one fixed point in X_+. It satisfies $0 \le v \le b$, $u_0 \le u \le 2 + b$, and is a smooth solution of (1.6) with the full derivative decay stated in the source. For every $z \in X_+$,*

$$\|v - z\|_\infty \le \frac{\|z - Tz\|_\infty}{1 - q}. \tag{4.3}$$

If $v_0 = 0$, $v_{n+1} = Tv_n$, then

$$\|v - v_n\|_\infty \le \frac{q^n b}{1 - q}, \qquad v_{2j} \le v \le v_{2j+1} \quad (j \ge 0). \tag{4.4}$$

Proof. For $v, w \ge 0$, all arguments $u_0 + v$ are at least 1. The scalar mean-value theorem bounds the difference of the source by $(7p + 3\ell)\|v - w\|_\infty$. Theorem 3.2 gives $\|Tv - Tw\| \le q\|v - w\|$ and $0 \le Tv \le b$. The complete cone is invariant, so Banach's theorem applies without assuming it is a vector space. The fixed point is C^1 by Lemma 3.1. Localizing the source near a point and integrating all but one derivative onto a compactly supported source factor gives one additional derivative whenever that source is C^j. Induction using smooth p, ℓ and $u \ge 1$ proves smoothness; rescaled cutoffs on balls of radius comparable to r give $|D^k v| \le C_k s^2 L^{-1} r^{-1-k}$ exactly as in the audited bootstrap. Source separation makes the boundary smooth. Rotation equivariance and uniqueness imply axisymmetry. Adding and subtracting Tz proves (4.3). Summing $\|v_{n+1} - v_n\| \le q^n \|v_1\| \le q^n b$ proves the first bound in (4.4). Since the powers are decreasing and the kernel positive, T reverses order. Starting with $0 \le v \le T0$ and applying T repeatedly proves the displayed alternating bracket; the even sequence increases and the odd sequence decreases. □

No efficiency comparison follows merely from the geometric rate. The cost of integrating the singular kernel, controlling all $x \in \Omega$, and bounding its tail remains part of an implementation.

4.2 A mass-sensitive weighted space

Define the Banach space

$$X_1 = \{v \in C(\Omega) : \|v\|_1 = \sup_{x \in \Omega} r|v(x)| < \infty\}.$$

Multiplication by r is an isometry onto $C_b(\Omega)$. The shell formula also gives the exact weighted operator bound

$$C_k(L) = \sup_{r \ge 1} r\mathcal{R}(r^{-k}\mathbf{1}_{r \ge L}) = \frac{L^{3-k}}{k-3} + \frac{L^{2-k}}{k-2}. \tag{4.5}$$

Indeed, on $[1, L]$ the product is a positive constant times $r + 1$; beyond L it increases to the displayed limit by (3.3). Positivity and continuous approximation give the operator bound on the continuous source space as before. On the nonnegative cone of X_1,

$$\|Tv\|_1 \le PC_6(L) + EC_4(L), \qquad \|Tv - Tw\|_1 \le q_1\|v - w\|_1, \quad q_1 = 7PC_7(L) + 3EC_5(L). \tag{4.6}$$

The extra source factor r^{-1} comes from $|v(y) - w(y)| \le \|v - w\|_1/|y|$. Thus every cone residual and iteration certificate remains valid in X_1 with $q_1 < 1$, and the stronger state conclusion is $|v - z| \le e_1/r$. Neither a change in the PDE nor an artificial truncation is involved.

4.3 Positive ball and complete derivative

On the open set $U = \{z \in X : \inf_\Omega(u_0 + z) > 0\}$ put $F(z) = z - Tz$. For a ball $\overline{B}(z, \rho) \subset U$ write $a_0 = \inf(u_0 + z) - \rho > 0$ and

$$q_{a_0} = \frac{7Pa_0^{-8}}{2L^4} + \frac{3Ea_0^{-4}}{L^2}. \tag{4.7}$$

If $q_{a_0} < 1$ and $\|F(z)\| + q_{a_0}\rho \le \rho$, the ball contains a unique zero and its distance to z is at most $\|F(z)\|/(1 - q_{a_0})$. This follows by the same invariant-ball proof, now permitting a candidate which is not nonnegative; its fixed point is automatically nonnegative by positivity of T.

The actual derivatives are

$$DF(z)w = w + \mathcal{R}\{[7p(u_0 + z)^{-8} + 3\ell(u_0 + z)^{-4}]w\}, \tag{4.8}$$
$$D^2T(z)[w_1, w_2] = \mathcal{R}\{[56p(u_0 + z)^{-9} + 12\ell(u_0 + z)^{-5}]w_1w_2\}.$$

They are bounded Fréchet derivatives by the weighted majorants and Taylor remainders on positive intervals. The derivative Lipschitz constant on that ball is at most

$$K_{a_0} = \frac{28Pa_0^{-9}}{L^4} + \frac{12Ea_0^{-5}}{L^2}. \tag{4.9}$$

The source state is $u(y)$, not $u(x)$: there is no hidden observation state multiplier, no state-dependent normalization, and no moving-domain shape term in this fixed-L problem. If a bounded isomorphism $B : X \to X$ is supplied, a stronger preconditioned test is

$$\eta = \|BF(z)\|, \quad \kappa = \|I - BDF(z)\|, \quad \eta + (\kappa - 1)\rho + \|B\|K_{a_0}\rho^2 \le 0, \quad \kappa + \|B\|K_{a_0}\rho < 1.$$

This is a valid sufficient residual-ball test, not a claim that a finite matrix inverse is already an isomorphism of X.

4.4 Actual private Ram passages and their scope

The following are actual passages read, not routing-map substitutions. Paths are local and confidential. Each path is relative to `C:/Users/mural/OneDrive/Documents/RamNonlin-IE-PDE-astra/`.

1. `TR1-lin-nonlin/stability.tex`: "Nonlinear differentiation and certified solution neighborhoods", `thm:ball`, `eq:radii`, and "Transporting equations through Ram representations", `prop:transfer`. The residual here is $F = I - T$, with $X = Y = C_b(\Omega)$ and $B = I$ for the direct certificate. The actual weighted exterior estimates replace any implicit bounded-domain kernel bound. The "Weak Ram responses" formulas `eq:weak` and `eq:boundaryweak` warn against discarding the Robin boundary term. `thm:pert` concerns finite least-squares conditioning; it is not an infinite-dimensional PDE inverse theorem.

2. `TR6-lin-non-lin/TR6-lin-non-lin.tex`: "Unbounded domains and weighted sources", "Exact Localization and Its Analytical Limits", "Full Urysohn variation" (`eq:du`, `eq:ddu`), `eq:newtonfull`, `sec:shape`, and "Newton Solves and Quantitative Local Validation". The actual specialization is a source-only Hammerstein nonlinearity with negative powers. The shift $y = x - a$ gives

$$Tv(x) = \int_{x-\Omega} G_R(x, x-a)\{p(x-a)(u_0+v)(x-a)^{-7} + \ell(x-a)(u_0+v)(x-a)^{-3}\}\,\mathrm{d}a. \tag{4.10}$$

   The absolute Jacobian is 1. The singularity at $a = 0$, reflected boundary, noncompact source, and source measure are unchanged. This is an exact Ram representation, not a support reduction or a new existence theorem.

3. `TR8-lin-non-lin/TR8-lin-non-lin.tex`. The passages read are `sec:elliptic`, `thm:coercive`, `eq:operator-domain`, `eq:compatibility`, `thm:green-equivalence`, `thm:heat-equivalence`, and `sec:semilinear`, together with "Reflection parity is conditional" and "A local a posteriori existence certificate". Its bounded Lipschitz Dirichlet inverse $H^{-1} \to H_0^1$ is *not* our $\mathcal{R}$. Its positive Robin coefficient argument cannot be imported, since our outward-normal coefficient is negative. The retained lesson is domain-first equivalence; Lemma 3.1 supplies the exterior replacement. The heat equivalence was inspected but supplies no time evolution or parabolic conclusion here.

The concepts of Kelvin reflection, Newton potential, Banach contraction, energy uniqueness, and residual amplification are classical. The private Ram contribution used here is a compatible representation and explicit validation contract, not historical ownership of those methods. A bounded Ram change of coordinates C transports F to SFC^{-1} only after bounded isomorphism estimates for C, S are proved. No such new transform or historical priority claim is established here.

5 Tail, Perturbation, and Output Error Bounds

5.1 Implementation error is not the displayed residual alone

Suppose $z \geq 0$ is a continuous reconstruction and an implemented $\widehat{T}z$ has a proved uniform error $\|Tz - \widehat{T}z\| \leq \epsilon$. Theorem 4.1 gives

$$\|v - z\| \leq \frac{\|z - \widehat{T}z\| + \epsilon}{1 - q}. \tag{5.1}$$

Here ϵ must include quadrature, reconstruction, kernel, roundoff, and source truncation errors in one common sup norm. A finite grid maximum is not automatically a bound on this norm. To certify it one needs interval cell bounds or a derivative-based fill-distance estimate, including the unbounded exterior.

Proposition 5.1 (Exterior truncation and coefficient changes)**.** *For every $R \geq L$, omitting sources on $|y| > R$ incurs the uniform operator error*

$$\epsilon_{\text{tail}} \leq \frac{P}{2R^4} + \frac{E}{R^2} \tag{5.2}$$

on the whole nonnegative cone. If T_R is the truncated map, its unique cone fixed point v_R exists when $q < 1$, and $\|v - v_R\| \leq \epsilon_{\text{tail}}/(1 - q)$. For nonnegative smooth coefficient pairs with the same support radius, if $|p - \widehat{p}| \leq \delta_P r^{-6}$ and $|\ell - \widehat{\ell}| \leq \delta_E r^{-4}$, and both cone fixed points exist, then

$$\|v - \widehat{v}\| \leq \frac{\delta_P/(2L^4) + \delta_E/L^2}{1 - q}. \tag{5.3}$$

Proof. Apply Theorem 3.2 to the omitted source with radius R. For a hard truncation the source is merely measurable across R, but its potential is continuous and the same contraction acts on X_+; smoothness across R is not asserted. Its Lipschitz constant is at most q, so existence follows. Subtract the two fixed point equations and bound the difference of maps at the same state. Rearranging the term $q\|v - v_R\|$ gives the bound. The coefficient argument is identical; existence of the perturbed state is either separately certified by its own contraction constants or assumed as stated. □

Truncation changes the physical tail and potentially its angular momentum. The bound certifies an approximation to the scalar equation; it does not license deleting the electromagnetic angular tail in the physical construction or declaring its total J unchanged.

5.2 Mass and area enclosures

Define the positive linear source functional

$$\mathfrak{m}(H) = \frac{1}{2\pi}\int_\Omega (1 + r^{-1})H \, \mathrm{d}x, \quad M_k(L) = 2\left\{\frac{L^{3-k}}{k-3} + \frac{L^{2-k}}{k-2}\right\} \quad (k > 3). \tag{5.4}$$

An envelope $ar^{-k}\mathbf{1}_{r\geq L}$ has exactly $\mathfrak{m}$-value $aM_k(L)$. Hence for a candidate $z \geq 0$ with $\|v-z\| \leq e$, define the source-based approximations

$$H_z = p(u_0+z)^{-7} + \ell(u_0+z)^{-3}, \quad m_z = 2 + \mathfrak{m}(H_z), \quad A_z = \int_{\mathbb{S}^2} (2 + z(1,\omega))^4 \, \mathrm{d}A.$$

The number m_z is not asserted to be the ADM mass of the off-equation metric $(u_0+z)^4\delta$. It is the source functional used to approximate the exact on-equation mass. If $2 + \max(b, \|z\|) \leq U$, then

$$|m - m_z| \leq [7PM_6(L) + 3EM_4(L)]e =: e_m, \qquad |A - A_z| \leq 16\pi U^3 e =: e_A. \tag{5.5}$$

The proof is the pointwise negative-power difference bound followed by integration for mass, and the mean-value inequality $|a^4 - b^4| \leq 4U^3|a-b|$ integrated over a sphere of area 4π. Additional certified quadrature errors for m_z, A_z must be added to e_m, e_A. The direct omitted-tail mass error is at most $PM_6(R) + EM_4(R)$, of order R^{-1}, even though the potential tail error is of order R^{-2}.

The weighted certificate gives a sharper mass-focused alternative:

$$|m - m_z| \leq 2q_1\|v - z\|_1, \qquad \|v - v_R\|_1 \leq \frac{PC_6(R) + EC_4(R)}{1 - q_1}. \tag{5.6}$$

For the first inequality multiply the source Lipschitz envelope by $\|v-z\|_1/r$, then use $M_k = 2C_k$ in (5.4). For the second apply (4.5) to the omitted source, then subtract the two fixed-point equations. Both statements require nonnegative states and $q_1 < 1$ where a denominator is used.

For exact $J = -4s^2/15$, suppose $m, m_z \leq M_*$ and $A, A_z \geq A_* > 0$. Then

$$|D - D_z| \leq 2M_* e_m + \left(\frac{1}{16\pi} + \frac{4\pi J^2}{A_*^2}\right) e_A. \tag{5.7}$$

This follows by differentiating m^2 and $-A/(16\pi) - 4\pi J^2/A$ along the interval between the two output pairs and taking absolute values. A strictly negative upper bound proves the bare scalar defect sign. To conclude a geometric counterexample one must additionally retain the constraints, decay, outermostness, and all-cuts area properties audited in Section 2.

5.3 Tail logarithms and the safe mass calculation

The first moment of an r^{-4} source in dimension three generally diverges logarithmically. For $r = |x| \geq 2L$, $|H| \leq ar^{-4}$, put $M_H = (4\pi)^{-1}\int H$. Splitting the Newton integral into $|y| \leq r/2$ and its complement gives

$$\left|\mathcal{N}H(x) - \frac{M_H}{r}\right| \leq \frac{a}{r^2}\{4\log(r/L) + 10\}. \tag{5.8}$$

Here is a direct proof. On the inner region the mean-value estimate for the Newton kernel is at most $4|y|/r^2$; integrating its envelope yields $4a\log(r/L)/r^2$. On the outer region, first restrict to $|x-y| \leq r/2$. There $|H| \leq 16a/r^4$ and the integral of $|x-y|^{-1}/(4\pi)$ over that ball is $r^2/8$, giving $2a/r^2$. On the remainder the kernel is at most $2/r$, whereas $(4\pi)^{-1}\int_{|y|>r/2}|H| \leq 2a/r$, giving $4a/r^2$. The subtracted monopole over this outer region contributes at most $2a/r^2$. These bounds are below the stated safe constant 10.

The reflected term equals $\mathcal{N}H(0)/r + O(r^{-2})$ because the ordinary potential is smooth near zero. Thus v admits a leading monopole, but a blanket $O(r^{-2})$ remainder for the ordinary potential does not follow merely from the source envelope. To see the logarithmic mechanism, write $H = r^{-4}a_1(\omega)$ with a_1 a degree-one spherical harmonic in a far region. Since

$$\Delta(r^\alpha Y_l) = \{\alpha(\alpha+1) - l(l+1)\}r^{\alpha-2}Y_l,$$

differentiation in α at $\alpha = -2, l = 1$ gives $-\Delta(r^{-2}\log r\, Y_1) = 3r^{-4}Y_1$. A positive angular envelope plus a small dipole can therefore retain positivity and still produce a logarithmic dipole term. The actual manuscript coefficients are even under $x \mapsto -x$, and uniqueness forces u and H to be even; their degree-one component vanishes. Formula (2.2), based on exact flux, is valid independently of this cancellation. There is nonetheless an actual logarithm at the next resonant order, as the independent review suggested and the following argument establishes.

Theorem 5.2 (A nonzero subleading quadrupole logarithm)**.** *For a solution in this family with $s \neq 0$, put $c_\infty = m/2$ and $P_2(t) = (3t^2-1)/2$. Its quadrupole coefficient $v_2(r) = (5/2)\int_{-1}^{1} v(r,t)P_2(t)\,\mathrm{d}t$ satisfies*

$$v_2(r) = \frac{5s^2}{21}r^{-2} - \frac{4c_\infty s^2}{7}r^{-3}\log r + O(r^{-3}). \tag{5.9}$$

This is consistent with every decay estimate in the source.

Proof. Pair y and $-y$ in the Newton integral. For $|y| \le r/2$, the linear Taylor terms in the two kernels cancel, leaving an error at most $C|y|^2/r^3$. Its integral against $|H|$ is $O(r^{-2})$, since $\int_{L\le|y|\le r/2}|y|^2|H|\,\mathrm{d}y = O(r)$. The outer region is $O(r^{-2})$ by the same split as in (5.8). The image potential has no linear Taylor term at zero, by evenness. It follows that $u = 1 + c_\infty/r + O(r^{-2})$ with $c_\infty = 1 + (4\pi)^{-1}\int H + \mathcal{N}H(0) = m/2$.

For $r \ge 2L$, $\ell = r^{-4}\ell_4(t)$, and exact Legendre decomposition gives

$$\ell_4 = s^2\left(\frac{8}{15}P_0 + \frac{20}{21}P_2 + \frac{18}{35}P_4\right).$$

Taylor expansion of the scalar negative power, now justified by the uniform monopole remainder, gives $H = r^{-4}\ell_4 - 3c_\infty r^{-5}\ell_4 + O(r^{-6})$. Since the solution is smooth and axisymmetric, projection of its PDE onto P_2 is legitimate and yields

$$-v_2'' - 2r^{-1}v_2' + 6r^{-2}v_2 = \frac{20s^2}{21}r^{-4} - \frac{20c_\infty s^2}{7}r^{-5} + O(r^{-6}).$$

Direct differentiation gives $-\Delta(r^{-2}P_2) = 4r^{-4}P_2$ and $-\Delta(r^{-3}\log r\, P_2) = 5r^{-5}P_2$. Subtract the two displayed particular solutions in (5.9); the remainder w_2 obeys the same radial operator with forcing $f_2 = O(r^{-6})$ and decays at infinity. Variation of constants, for any fixed $R_0 \ge 2L$, writes it as

$$w_2(r) = A_0 r^{-3} + \frac{1}{5}\left[r^{-3}\int_{R_0}^{r}\rho^4 f_2(\rho)\,\mathrm{d}\rho + r^2\int_r^\infty \rho^{-1}f_2(\rho)\,\mathrm{d}\rho\right].$$

The first integral converges to a finite constant with $O(r^{-1})$ tail, and the second contribution is $O(r^{-4})$. No growing r^2 homogeneous mode is allowed by decay. Thus $w_2 = O(r^{-3})$, proving the expansion. Positivity gives $m \ge 2$, so its logarithmic coefficient is nonzero when $s \neq 0$. Finally derivatives of $r^{-3}\log r$ satisfy the source's much weaker $O(r^{-1-k})$ bounds; there is no contradiction. □

6 Scalar Solvability Without Smallness

The following statements extend the scalar domain to $L > 1$ and every finite real s, with the same smooth cutoff and coefficient formulas. The inverse proof and shell formula require only $L > 1$; the restriction $L > 4$ was used for the earlier convenient uniform derivative constants. For fixed s, L, local regularity estimates may have constants depending on them. No uniform geometric smallness is asserted in this extension.

Theorem 6.1 (Every finite amplitude has a scalar solution)**.** *For every $s \in \mathbb{R}$ and $L > 1$, with any fixed smooth cutoff as specified, (1.6) has a smooth positive solution with $u - u_0 = O(r^{-1})$. It obeys*

$$0 \le u - u_0 \le b, \qquad \|u - u_0\|_1 \le PC_6(L) + EC_4(L),$$

where P, E, b retain their definitions, without requiring $q < 1$.

Proof. The closed convex bounded set $\mathcal{K} = \{v \in C_b(\Omega) : 0 \le v \le b\}$ is nonempty. The exact majorants prove $T(\mathcal{K}) \subset \mathcal{K}$ for any finite s, since $u_0 + v \ge 1$. The finite Lipschitz estimate $\|Tv - Tw\| \le q\|v - w\|$ proves continuity; it need not have $q < 1$. On each compact exterior set the potential-gradient integrals are uniformly bounded by the common envelope $p + \ell$. Local integrability of $|x - y|^{-2}$ handles its singularity, and near the boundary the source is separated by the positive distance $L - 1$. Thus $T(\mathcal{K})$ is locally equicontinuous. Its tail is uniformly bounded by $[PC_6(L) + EC_4(L)]/r$.

Given a sequence in the image, Arzela–Ascoli on increasing compact exteriors and a diagonal choice produce a subsequence converging uniformly on every compact set. To obtain uniform convergence on all of Ω, choose a large radius so that the common tail bound is less than any prescribed tolerance; inside that radius use compact convergence. The image is therefore relatively compact in C_b. The compact-map Schauder fixed-point theorem on a closed bounded convex set yields a fixed point. This classical theorem is recorded, for example, in Holst–Nagy–Tsogtgerel [5], Appendix A.1, Theorem 9; no geometric theorem from that paper is imported. The same localized one-derivative bootstrap as in Theorem 4.1 gives smoothness and derivative decay, now with constants depending on s, L. The inverse gives the exact Robin trace and the limit at infinity. Positivity and the two bounds follow from the majorants at the fixed point. □

Theorem 6.2 (Positive-solution uniqueness on the exterior)**.** *Let smooth nonnegative p, ℓ vanish for $r < L$, with $p = O(r^{-6})$, $\ell = O(r^{-4})$ and the corresponding smooth coefficient bounds. There is at most one smooth positive solution of (1.6) satisfying $u - u_0 = O(r^{-1})$. No smallness condition on p, ℓ is needed for this uniqueness statement. Every such solution has $u \ge u_0$.*

Proof. A positive smooth solution tending to 1 has a uniform positive lower bound: use compactness on a bounded exterior and the limit on its complement. Its source thus belongs to $Y_{4,L}$. Subtracting $\mathcal{R}H$ from $u - u_0$ and applying Lemma 3.1 proves $u - u_0 = \mathcal{R}H \ge 0$. In particular $\nabla(u - u_0) = O(r^{-2})$ by the potential estimate, independently of a contraction hypothesis.

For two solutions let $w = u_1 - u_2$. The negative powers give

$$-\Delta w = -c(x)w, \quad c = \int_0^1 \{7p[u_2 + t(u_1 - u_2)]^{-8} + 3\ell[u_2 + t(u_1 - u_2)]^{-4}\}\,\mathrm{d}t \ge 0.$$

We have $w = O(r^{-1})$, $\nabla w = O(r^{-2})$, and $c = O(r^{-4})$. Integrate over $1 \le r \le R$ and let $R \to \infty$. The outer boundary term tends to zero, all integrals converge, and the inner outward normal derivative is $\partial_n w = w/2$. Therefore

$$\int_\Omega |\nabla w|^2\,\mathrm{d}x - \frac{1}{2}\int_S w^2\,\mathrm{d}A + \int_\Omega cw^2\,\mathrm{d}x = 0. \tag{6.1}$$

On each ray, $w(1, \omega) = -\int_1^\infty w_r\,\mathrm{d}r$, and Cauchy–Schwarz with $\int_1^\infty r^{-2}\,\mathrm{d}r = 1$ gives

$$\int_S w^2\,\mathrm{d}A \le \int_{\mathbb{S}^2}\int_1^\infty r^2|w_r|^2\,\mathrm{d}r\,\mathrm{d}A \le \int_\Omega |\nabla w|^2\,\mathrm{d}x.$$

Thus the left side of (6.1) is at least $\frac{1}{2}\int|\nabla w|^2 + \int cw^2$, forcing $w = 0$ by decay. □

This strengthens the stated cone-and-smallness uniqueness route in the source, and rules out a second smooth positive decaying branch even if the explicit Picard criterion fails. Theorem 6.1 now supplies existence at every finite amplitude, but neither argument establishes convergence of Picard iteration when its contraction test fails. The trace-energy inequality, not a bounded-Dirichlet coercivity theorem, is what controls the negative outward Robin coefficient.

Corollary 6.3 (Scalar continuation on bounded amplitude intervals)**.** *For fixed $L > 1$, the unique scalar solution depends continuously on $s \in \mathbb{R}$ in C_b. Its mass source functional and boundary area are continuous. Axisymmetry and antipodal evenness hold at every finite amplitude.*

Proof. On any bounded amplitude interval the preceding proof supplies common majorants, equicontinuity, and uniformly vanishing tails. For $s_n \to s$, the solutions therefore have a uniformly convergent subsequence. Continuity of the map in s and its state, using the same majorants, identifies the limit as a fixed point at s. Uniqueness makes every subsequential limit the same, hence the whole sequence converges. Dominated convergence with weights 1 and r^{-1} proves mass continuity; uniform boundary convergence proves area continuity. Equivariance under rotation and antipodal reflection together with uniqueness proves the symmetries. □

This is a scalar parameter extension, not an assertion of outermostness, all-cuts minimality, or violation of (1.4) at arbitrary amplitude.

7 Explicit Barrier and Finite-Parameter Certificate

7.1 Derivative constants without differentiating the source

For $|H| \le ar^{-4}\mathbf{1}_{r\ge L}$ with $L > 4$, direct Newton kernel estimates give

$$\|\mathcal{R}H\| \le aL^{-2}, \quad |r\partial_r \mathcal{R}H| \le 41aL^{-2}, \tag{7.1}$$

and on $1 \le r \le 2$,

$$|\partial_r \mathcal{R}H| \le \frac{7}{6}aL^{-2}, \quad |\partial_r^2 \mathcal{R}H| \le \frac{17}{6}aL^{-2}. \tag{7.2}$$

To verify the constants, write $N = \mathcal{N}H$. On $|x| \le L/2$, $|\nabla N| \le 4a/(3L^3)$ and every directional second derivative is at most $4a/L^4$, because $|x-y| \ge |y|/2$ and $\|D^2|z|^{-1}\| \le 2|z|^{-3}$. Also $|N| \le a/(2L^2)$ by the exact radial envelope. On $r \ge L/2$ split at $|x-y| = r/2$. The inner gradient part is at most $8a/r^3$, the outer part at most $4a/(Lr^2)$, hence $r|\nabla N| \le 40a/L^2$. On the inner region $r|\nabla N| \le 2a/(3L^2)$. For $r^{-1}n(1/r)$ the first derivative is $-r^{-2}n - r^{-3}n'$ and the second is $2r^{-3}n + 4r^{-4}n' + r^{-5}n''$. At its argument $1/r \le 1$ the separated estimates apply. For the first reflected derivative multiplied by r, its bound is $a/(2L^2) + 4a/(3L^3) < 5a/(6L^2)$, giving 41. Adding the ordinary and reflected derivatives on $[1,2]$ gives $a/(2L^2) + 8a/(3L^3) \le 7a/(6L^2)$ and $a/L^2 + 16a/(3L^3) + 8a/L^4 \le 17a/(6L^2)$.

Proposition 7.1 (A quantitative sphere barrier)**.** *For the fixed point of Theorem 4.1, put $a = E + P/L^2$ and $\gamma = a/L^2$. If*

$$L > 4, \qquad \gamma \le 1/332, \tag{7.3}$$

then contraction holds and the source's radial barrier satisfies

$$W \ge (r-1)/8 \quad (1 \le r \le 2), \qquad W \ge 1/4 \quad (r \ge 2).$$

Proof. The nonlinear source is bounded by ar^{-4} since $u \ge 1$. Also $q \le 7\gamma < 1$. For $r \ge 2$ use $W = 1 - r^{-1} + v + 2rv_r \ge 1/2 - 83\gamma \ge 1/4$. On $[1,2]$, (7.2) gives $W_r = r^{-2} + 3v_r + 2rv_{rr} \ge 1/4 - (89/6)\gamma \ge 1/4 - 15/332 > 1/8$. Since $W(1) = 0$, integration proves the remaining inequality. These constants are deliberately conservative. □

Retaining the smooth seed identities and the topological comparison arguments in Section 2, this explicit scalar criterion is sufficient for their outermostness and all-enclosing-cuts conclusions. It is stronger information than merely reducing the unknown constant in a contraction estimate: it closes a quantitative link to geometry.

7.2 A conservative finite witness

For the manuscript's displayed cutoff, let $x = z - 1 \in (0, 1)$ and

$$\chi(z) = \frac{e^{-1/x}}{e^{-1/x} + e^{-1/(1-x)}}.$$

Put $z_0 = 2x - 1$. Direct differentiation of the logistic form gives

$$\chi'(z) = \frac{2(1 + z_0^2)}{(1 - z_0^2)^2} \operatorname{sech}^2\left(\frac{2z_0}{1 - z_0^2}\right) \le \frac{2}{1 + z_0^2} \le 2.$$

For the inequality use $\cosh^2 w \ge 1 + w^2$ and $(1 - z_0^2)^2 + 4z_0^2 = (1 + z_0^2)^2$. Endpoint flatness supplies the derivative bound at the two endpoints as well. Since $1 \le z \le 2$, this proves $d_\chi \le 4$, hence the sharper constants

$$P = s^4/4, \quad E = 6s^2, \quad L = 20000. \tag{7.4}$$

No sampling of the cutoff derivative is used for this bound. In particular $L \ge 50$ already suffices for (7.3), because $(6 + 1/(4 \cdot 50^2))/50^2 < 1/332$ and this quantity decreases with L. This reproduces the independent reader's $L = 50$ geometric threshold using our slightly different conservative derivative estimates. For comparison, the explicit sup-norm and weighted contraction constants at $s = 1$ obey

$$q(7) = 1009/2744 < 1/2, \qquad q_1(5) = 102203/250000 < 1/2.$$

These smaller radii certify scalar contraction, not the geometric barrier.

Theorem 7.2 (Finite scalar and defect certificate)**.** *For (7.4), uniformly over $s \in [0, 1]$, the scalar solution exists, is positive and unique, and satisfies Proposition 7.1. At $s = 1$, using the source output identities (2.2) and $J = -4/15$, its bare defect is strictly negative. This conclusion needs no numerical evaluation of the fixed point or its singular integral.*

Proof. Here $\gamma \le (6 + 1/(4L^2))/L^2 < 1/332$ by rational arithmetic. The preceding results supply the scalar conclusions and barrier. Let

$$B = PM_6(L) + EM_4(L).$$

Since $H \le p + \ell$, the exact mass identity gives $2 \le m \le 2 + B$. Since $0 \le v \le b$, the exact boundary area obeys $64\pi \le A \le 4\pi(2 + b)^4$. Hence

$$D(1, L) \le (2 + B)^2 - 4 - \frac{16}{225(2 + b)^4}. \tag{7.5}$$

At $L = 20000$, $P = 1/4$, $E = 6$, one has $B < 601/10^6$ and $b < 16/10^9$. Thus $(2 + B)^2 - 4 < 241/10^5$, while $16/[225(2 + b)^4] > 441/10^5$. The right side of (7.5) is less than $-1/500 < 0$. All inequalities involve rationals; the companion check verifies both these weaker bounds and the direct rational expression. The proof uses no decimal integration or observed convergence. □

The finite witness is intentionally large and not an optimized scale. It is a quantitative specialization of the pinned construction, not a different headline counterexample or evidence of historical novelty. The source's seed-to-geometry identities remain explicitly inherited; the local certificate establishes the scalar, barrier, and defect inequalities conditional on those identities.

For equality, an explicit version of the source argument also follows. Uniformly $b \le Cs^2$ with $C = 6/L^2 + 1/(8L^4)$ and $M = 2 + C$. Hence $A - 64\pi \le 16\pi M^3 C s^2$. The source tail lower bound gives

$$D(s, L) \ge s^2 \left\{ \frac{4}{3M^3 L} - M^3 C \right\} - \frac{s^4}{225}.$$

At the displayed L the brace is positive, as an exact-rational check confirms. Any sufficiently small nonzero s with $s^2 < 225\{4/(3M^3L) - M^3C\}$ therefore has positive defect. Uniform contraction makes $s \mapsto v_s$ continuous; weighted domination makes the mass continuous, and the boundary area is continuous in the sup norm. The intermediate value theorem then gives a nonzero equality amplitude, without locating or computing it. The non-realization argument is the audited source Maxwell-field obstruction, not an independently reconstructed spacetime rigidity proof.

For a concrete bracket one can instead retain the simpler $M = 3$ and write, at $L = 20000$,

$$D(s, L) \ge s^2\{4/(81L) - 27C\} - s^4/225.$$

At $s = 1/100$ its value is the strictly positive rational

$$\frac{167929599997813}{10368000000000000000000000}.$$

Together with Theorem 7.2, this puts at least one equality amplitude in $(0.01, 1)$ for the displayed cutoff. It does not prove uniqueness or compute the location of that root.

8 Public Comparison, Limits, and Research

The primary public comparison was checked on October 7, 2026 by public titles and topics only. Khuri–Weinstein [2], equation (1.12), includes an electromagnetic angular-momentum surface term; the subsequent discussion requires additional asymptotic conditions for its disappearance. Its two-ended mass–angular-momentum–charge theorem is not the same problem as a one-ended exterior with the present minimum enclosing area. Sign conventions must be matched before importing its constraint equations. Dain–Khuri–Weinstein–Yamada [3], equations (2.20) and (2.22), use Coulomb-leading electric and magnetic expansions when identifying fluxes at infinity. These differ from the nonconstant angular r^{-2} tails used here. Their paper also distinguishes upper and lower area statements.

A current primary preprint by Xu [4], version 4 dated January 2, 2026, advertises an angular-momentum Penrose inequality in vacuum. Only its bibliographic record, abstract, and version history were checked here; its proof is not used or independently endorsed. A vacuum claim cannot resolve the present electromagnetic boundary-term distinction. The search was targeted, not an exhaustive literature or priority survey.

Compared with the source's qualitative CL^{-2} contraction, the present certificate has the advantages of a declared operator norm, fully propagated tail error, an admissible positive-state neighborhood, quantified radial barriers, and directly testable defect signs. Compared with a generic Ram panel argument, it retains the exterior source and the exact boundary reflection. Compared with a brute-force finite-domain solve, it separates R^{-2} field error from the slower R^{-1} mass tail. Its disadvantages are conservative constants, a very large demonstration radius, no implemented interval singular quadrature, and no proved computational acceleration.

There are specific next steps. A validated singular quadrature and an exterior cellwise residual bound would turn (5.1) into an executed numerical PDE certificate at smaller L.

Spherical-harmonic tail integration could retain the physically essential angular tails while reducing mass error. A source-sensitive bound using $u_0(r)^{-8}$ and $u_0(r)^{-4}$ instead of 1 would sharpen q. Quantitative weighted derivative residuals could certify geometric barriers around nonexact reconstructed states, where a sup-norm ball alone is insufficient. Extending the domain from a sphere to a distorted boundary would require a new boundary inverse or a justified perturbation of it; the Kelvin formula cannot simply be reused. None of these future steps is credited in the contribution assessment.

9 Verification and Review Record

The companion author check completed 43 bounded deterministic assertions. It verifies the radial envelope formulas and their PDE/trace identities symbolically, the angular integrals for the gravitational and electromagnetic fluxes, the dipole logarithmic identity, and the finite-parameter inequalities using exact rational arithmetic. These are analytic-formula regression checks, not a numerical solution of the three-dimensional PDE, an interval proof of every source argument, or a performance benchmark.

The parent read the initial complete author report and independently approved its inverse norms, contraction and radii bounds, error propagation, trace-energy uniqueness, derivative constants, and finite-witness logic. The independent reader's complete proof notes and check results were then read and incorporated: weighted residuals, all-finite-amplitude scalar existence, the actual quadrupole logarithm, and the sharper cutoff and sign bounds. The parent separately reviewed and approved those added proofs, including their radial Green signs and exponents. This was an internal targeted mathematical review, not external peer review or blanket verification of the headline source theorem.

The native open request was queued. The parent's native compilation attempt returned `compile-failed`, with diagnostic

```
Unable to find standard directories for platform
```

and a trailing newline. No successful native compilation is claimed. The separate disk-export route is the unmodified `_tools/compile_report.py`, run with the bundled Python and installed MiKTeX. The artifact-specific disk compilation, contact-sheet inspection, detailed sample inspection, and final hash manifest are recorded in `_audit/M514-author*`. They are distinct from the native failure. No packages or TeX components were installed. The initial author-check attempt found SymPy unavailable; the successful check uses only exact standard-library rational, exponent, and polynomial arithmetic. Delivery and shared catalogue updates are controlled by the parent.

A Reproducible Input and Dependency Index

The study root is

```
C:/Users/mural/OneDrive/Documents/OpenAI-722-math/
```

Under the source root named in Section 1, the complete read set is `README.md`, `build/paper.tex`, `build/preamble.tex`, `build/references/references.bib`, and `build/sections/00-introduction.tex` through `build/sections/04-counterexamples.tex`. Their roles are, respectively, metadata, inclusion closure, definitions, references, problem class and theorem, seed/conformal construction, scalar inverse and bootstrap, horizon/all-cuts geometry, and asymptotic/equality proof. No absent appendix has been silently assumed read.

The private corpus passages are exactly those indexed in Section 4.4. Full-file hashes identify the versions even where only the explicitly listed sections were needed. The public bibliography

does not contain confidential corpus text. Machine-readable author records are under `_audit/M514-author*`; the final score is in `_audit/contribution-assessment-514.json`. Those records distinguish input hashes, executed checks, review status, compilation status, visual inspection, and final freeze. The parent owns delivery and shared catalogue/progress updates.

References

[1] OpenAI, *Electromagnetic tails and the Kerr–Newman Penrose inequality*, October 5, 2026, pinned manuscript, commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Pinned public source directory. Local source files were the mathematical evidence read.

[2] M. Khuri and G. Weinstein, *The positive mass theorem for multiple rotating charged black holes*, Calc. Var. PDE 55 (2016), article 33. arXiv:1502.06290v2, especially equation (1.12) and its following discussion.

[3] S. Dain, M. Khuri, G. Weinstein, and S. Yamada, *Lower bounds for the area of black holes in terms of mass, charge, and angular momentum*, Phys. Rev. D 88 (2013), 024048. arXiv:1306.4739v2, especially equations (2.19)–(2.27).

[4] D. Xu, *The Angular Momentum Penrose Inequality*, arXiv:2512.06918v4, January 2, 2026. Current primary preprint record checked; proof not used.

[5] M. Holst, G. Nagy, and G. Tsogtgerel, *Rough solutions of the Einstein constraints on closed manifolds without near-CMC conditions*, March 31, 2008, author-hosted arXiv:0712.0798v2, Appendix A.1, Theorem 9. Only the classical abstract Schauder theorem is used here.

[6] Private Ram corpus, *TR1-lin-nonlin*, `stability.tex`, specific sections and labels in Section 4.4; confidential local source.

[7] Private Ram corpus, *TR6-lin-non-lin*, `TR6-lin-non-lin.tex`, specific sections and labels in Section 4.4; confidential local source.

[8] Private Ram corpus, *TR8-lin-non-lin*, `TR8-lin-non-lin.tex`, specific sections and labels in Section 4.4; confidential local source.